



MINING DEALER BEST PRACTICE

C32 Oil Jet Reusability Determination Fixture

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

C32 Oil Jet Reusability Determination Fixture.....	0
1.0 Introduction.....	1
2.0 Best Practice Description.....	1
3.0 Implementation Steps.....	2
4.0 Benefits.....	2
5.0 Resources Required.....	2
6.0 Supporting Attachments / References.....	2
7.0 Related Best Practices.....	3
8.0 Acknowledgements.....	3

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices.” Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

March 2015
1502-4.03-1369

1.0 Introduction

Borusan Makina Kazakhstan has made the fixture to easily check oil jets and make the right decision if it reusable or it should be replaced.

The wrong decision about the reusability of oil jets may lead to the incorrect angle of supplying the grease and cooling the piston head, excessive wear of the liner wall and piston skirt, the piston head overheating and finally to the emergency shutdown of the engine. That's why it is very important to prevent the reuse of the damaged oil jets.

During the visual inspection of the injectors it is almost impossible to determine the bent or damaged oil jets with the naked eye. This fixture helps us to see if the oil jet is OK or it is damaged.

2.0 Best Practice Description

This fixture is made of the polyester resin. A new injector has been used to mold the correct form of the jet.



C32 Oil Jet Reusability Determination Fixture

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

C32 Oil Jet Reusability Determination Fixture

DATE
3/20/2015

CHG
NO
00

NUMBER
1502-4.03-1369

3.0 Implementation Steps

The fixture is in use.

It is simple to use this fixture. If the oil jet inspected does not match the pattern, it cannot be reused and should be replaced. If it matches the pattern, this oil jet can be reused.



The oil jet matches the fixture



The oil jet does not match the fixture

4.0 Benefits

- Increased defect list quality.
- Reduced defect list preparation time.
- Prevention of the troubles that occur if the overlooked damaged oil jet has been reused.

5.0 Resources Required

- Polyester resin and the new injector in order to make the pattern.

6.0 Supporting Attachments / References

N/A

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

C32 Oil Jet Reusability Determination Fixture

DATE
3/20/2015

CHG
NO
00

NUMBER
1502-4.03-1369

7.0 Related Best Practices

N/A

8.0 Acknowledgements

The idea of creating & implementing this Best Practice belongs to:

Ali CEKIC
Engine Team Leader
Borusan Makina Kazakhstan LLP
acekic@borusan.com
+7 (701) 951 97 97

Yerbol AITPAYEV
Service Engineer
Borusan Makina Kazakhstan LLP
yaitpayev@borusan.com
+ 7 (771) 050 69 69

Sergey KOROTKIKH
Maintenance personnel
Borusan Makina Kazakhstan LLP

This Best Practice was written by:
Gulnaz MANSUROVA
CRC Process Engineer
Borusan Makina Kazakhstan LLP
gmansurova@borusan.com
+7 (701) 951 98 14

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

C32 Oil Jet Reusability Determination Fixture

DATE
3/20/2015

CHG
NO
00

NUMBER
1502-4.03-1369